

Technology Stack

Date	25 June 2026
Team ID	SWTID-2026-2030
Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	2 Marks

Define Your Technology Stack:

The Voice-Based Concept Understanding Analyzer (VBCUA) was developed using Python along with modern AI and speech-processing libraries to automate the evaluation of students' spoken conceptual explanations. The application is built using FastAPI for backend services and Streamlit for developing an interactive web interface. Visual Studio Code was used as the integrated development environment (IDE) for coding, debugging, and project management. The system processes audio recordings in the .wav format, while FFmpeg is used for audio conversion and preprocessing to ensure compatibility during speech analysis.

The application utilizes OpenAI Whisper for accurate speech-to-text conversion, Librosa for audio feature extraction and analysis, and Sentence-BERT (SBERT) for semantic similarity analysis between the student's explanation and the reference concept. ReportLab is used to generate detailed PDF evaluation reports containing transcripts, scores, and AI-generated feedback. Together, these technologies provide an efficient, accurate, and user-friendly solution for automated voice-based concept evaluation and personalized learning feedback.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Streamlit, HTML, CSS	Provides a simple, responsive, and interactive web interface for students to record voice explanations, view AI evaluation results, audio visualizations, and download reports.
2	Backend / Server-Side	Python, Flask	Handles speech processing, AI model execution, application logic, report generation, and communication between the frontend and database efficiently.
3	Database / Data Storage	MySQL	Stores user accounts, speech transcripts, evaluation scores, feedback history, and generated reports securely with efficient relational data management.
4	Cloud / Hosting	Localhost / Streamlit	Used for local development and testing,

	/ Deployment	Cloud (Future Deployment)	with the option to deploy the application online using Streamlit Cloud for easy access and scalability.
5	Version Control & CI/CD	Git, GitHub	Enables version control, collaboration among team members, code backup, and project management throughout development.
6	Third-Party APIs / Other Tools	Google Speech Recognition API, OpenAI API/Gemini API (AI Evaluation), Matplotlib, ReportLab, FFmpeg	Speech Recognition API converts speech to text, AI API evaluates conceptual understanding, Matplotlib generates audio/performance visualizations, ReportLab creates PDF reports, and FFmpeg processes audio files.